Multi-Agent Reinforcement Learning for URLLC Modern Random Access

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***Abstract*—We tackle the problem of scheduling Ultra-Reliable Low-Latency Communications (URLLC) on the uplink by relying on Modern Random Access (MRA) and Multi-Agent Reinforcement Learning (MARL). We first formulate a URLLC MRA (uMRA) problem with a Decentralized Partially Observable Markov Decision Process (Dec-POMDP). We then propose uMRA-HAPPO, a MARL-based solution employing the Heterogeneous Agents Proximal Policy Optimization (HAPPO) algorithm. Finally, we assess the performance of our proposed algorithm in scenarios inspired by 3GPP use cases, showing higher reliability than deterministic and random traditional strategies under a strict latency constraint.**

***Index Terms*—URLLC, uplink scheduling, Multiple Access, Modern Random Access, MARL, Dec-POMDP, HAPPO**

I. Introduction

Ultra Reliable Low Latency Communications (URLLC) were set as an objective by the Third Generation Partnership Project (3GPP) in 5th-generation (5G) systems for supporting emerging mission critical applications such as telemedicine, intelligent transportation and industrial automation [1]. Various latency and reliability requirements were defined depending on the use cases, with a general requirement specifying the transmission of a 32-Byte packet within a 1ms deadline with a success probability of 99.999% [2]. However, the challenge remains to improve spectrum efficiency and to achieve even more stringent requirements for the 6th generation [3]. In particular, the stringent constraints of URLLC pose important challenges to Medium Access Control (MAC) protocols on the wireless channel. The uplink between User Equipment (UE) and the Base Station (BS) is particularly challenging as the information is distributed among the devices and can be centralized in the network only at the cost of a significant signalling load [4]. Deep Reinforcement Learning (DRL)-based approaches are seen as promising candidates to solve these challenges, as they naturally produce uplink scheduling policies capable of adapting to sporadic traffic and varying channel conditions [5]. On the other hand, Modern Random Access (MRA) schemes based on Slotted-ALOHA enhanced with the transmission of multiple packet replicas at the users side and Successive Interference Cancellation (SIC) at the BS [6] have shown great results for throughput maximization in uncoordinated multiple access [7]. Bringing together these two approaches, we propose a Multi-Agent RL (MARL)-based algorithm to address the URLLC uplink Multiple Access (MA) problem through the MRA framework.

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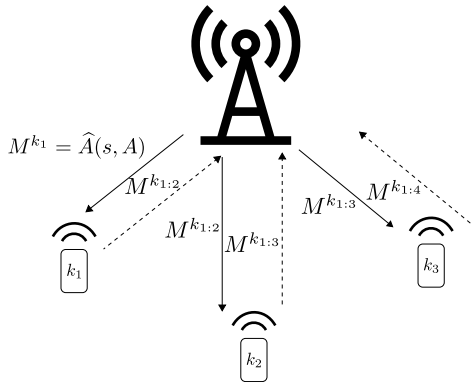


Fig. 1. System model: a network composed of 1 BS and several UEs. uMRA-HAPPO sequential update procedure description.

A. Related Work

Uplink MA schemes can be divided into two categories: Grant-Based (GB) and Grant-Free (GF). In GB schemes, for each of their transmissions, the UEs need to be granted a set of time-frequency resources by the BS. This procedure is part of the baseline protocol adopted in 5G NR with the Scheduling Request (SR) and Dynamic Scheduling (DS) mechanisms [8], [9]. This four-way handshake process is well-suited to sporadic traffic but very time consuming, making this approach incompatible with URLLC constraints. Some works, e.g. [5], [10], have explored DRL-based approaches with the BS anticipating the UEs needs for transmission to skip the SR step. However, the weakness of these approaches lies in the partial observability of the traffic information at the BS side. To solve these challenges, GF schemes have been seen as promising approaches. Following Wu et al. [11], these schemes can be separated into two groups: coordinated and uncoordinated access. In the former, UE transmissions are coordinated by the BS, which can e.g. pre-allocate uplink resources in a periodic manner. This is the case for the Semi-Persistent Scheduling (SPS) scheme introduced in LTE [12], and further enhanced in 5G NR with the Configured Grant (CG) allocation mechanism [9]. However, SPS becomes greatly resource inefficient when the traffic is sporadic, see [13] for a DRL approach studying the tradeoff between SPS and DS. In uncoordinated schemes, UEs operate uplink transmissions directly, without relying on BS grants. In particular, MRA achieves a much higher throughput than Slotted-ALOHA [6], [7]. However, MRA has never been considered in conjunction with URLLC, except in [14] where the protocol is not specifically optimized.

Authors of [15], [16] leverage DRL-based algorithms for MRA but ignore latency constraints. We hence propose a MRA DRL-based approach to URLLC uplink multiple access with strict deadlines. We base our scheme on MARL, as the uncoordinated context of MRA naturally matches a multi-agent setting. Prior works have investigated MARL for uplink URLLC, such as [17], [18], where authors tackle the multi-channel access problem. Other works, such as [19], have tackled uplink URLLC with MARL and SIC by adjusting the number of packet retransmissions, but to the best of our knowledge, no approach leveraging MARL in uplink URLLC has been proposed in the context of MRA with an explicit uplink slot choice behaviour optimization as we propose in this paper.

B. Contributions

Our contributions can be summarized as follows:

- We formulate the uMRA (URLLC MRA) problem as the planning problem of a Dec-POMDP.
- We introduce uMRA-HAPPO, a MARL algorithm based on HAPPO [20] and designed to numerically approach the solution of the uMRA problem.
- Through simulations, we show that uMRA-HAPPO outperforms traditional deterministic and random access schemes in 3GPP inspired scenarios. We also highlight an interesting tradeoff in the design of the frame length.

Notations: For $x \in \mathbb{R}$, $\lfloor x \rfloor$ and $\mathrm{frac}(x)$ are the floor and the fractional part of x, resp. Matrices (vectors) are in bold upper (lower) case. Subscript indices indicate time, while superscripts indicate vector or matrix components. $\mathcal{M}_{m\times n}(\mathbb{N})$ denotes the set of $m \times n$ matrices of positive integers. For $\boldsymbol{A} \in \mathcal{M}_{m\times n}(\mathbb{N})$, $\boldsymbol{A}^{k,:}$ denotes the k-th line of $\boldsymbol{A}$ and $\boldsymbol{A}^{:,k}$ its k-th column. The subsets of a family of p vectors or matrices $\boldsymbol{v}^{1:p} = \{\boldsymbol{v}^1, \boldsymbol{v}^2, \dots, \boldsymbol{v}^p\}$ are denoted by $\boldsymbol{v}^{k_{1:m}} = \{\boldsymbol{v}^{k_1}, \dots, \boldsymbol{v}^{k_m}\}$. The same notation is adopted for subscript indexes. The set of all probability distributions defined over a set S is denoted $\Delta(S)$.

II. System Model

A. Network model

We consider a wireless network of K heterogeneous UEs, each denoted by $k \in [\![1, K]\!]$ concurrently operating uplink transmissions to a single BS, as represented in Fig. 1. Time is divided into time-slotted radio frames, where each slot lasts for T_s seconds, see Fig. 2. Frames are numbered and indexed by $t \in \mathbb{N}$. At each frame t, each UE chooses a subset among m available time slots for its uplink transmission according to a certain policy, after which comes a single downlink feedback time slot. A guard slot accounts for signal propagation delays. A frame hence lasts for $T_f = T_s(m+2)$ seconds. Frequency bandwidth is divided into two parts: the uplink data part, which is dedicated to packet transmissions, and the reference signals part, which is dedicated to the transmission of uplink pilots. UE pilot signals are sent on

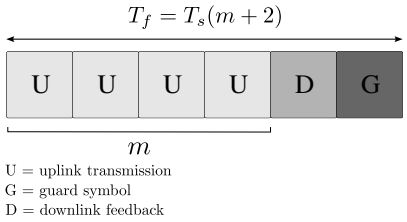


Fig. 2. Radio frame structure.

pre-allocated orthogonal sub-carriers in order to avoid intra-cell pilot contamination [21]. Our communication model is based on the MRA framework [22]. At the beginning of the frame, each UE selects a group of uplink slots during which it will transmit copies of its head-of-buffer packet to the BS. Each packet copy contains pointers (slot indices) to the slots containing other copies of the same packet, and is sent along with an uplink pilot. As it is assumed in the MRA literature, transmissions are subject to collisions but can be recovered thanks to SIC. A transmission received without collision is supposed to be successfully decoded. If multiple UEs transmit in the same slot, there is a collision. Transmissions can however still be successful if interfering copies can be decoded in other slots and removed with SIC. The uplink transmissions contained in the remaining collided signals at the end of this decoding procedure are lost. After decoding, the BS transmits a downlink feedback message to the UEs. This message contains per-UE ACK/NACKs and information required for each UE to learn its own policy.

B. Traffic model and buffer evolution

We adopt an aperiodic traffic model which is typical for 3GPP time-varying traffic URLLC scenarios [23]. Each UE k generates packets according to a Poisson process of rate λ_k. Every UE k has a strict latency constraint (deadline) of $\overline{\delta^k}$ seconds between its generation time in the buffer and the successful reception of an ACK; if a packet has not been successfully decoded by the BS before $\overline{\delta^k}$ seconds, it is discarded. If a transmission fails (due to a collision), the packet can be re-transmitted up until its deadline is met. We define $\delta^k := \lfloor \overline{\delta^k}/T_f \rfloor$, the deadline in number of frames, and we suppose that deadlines $\{\delta^k\}_{k=1\dots K}$ are known by the BS. We assume that the UEs have unlimited buffer capacity and that the packet queue operates on a "first come, first served" principle. For any UE k, we define the buffer status at t by the vector $\boldsymbol{b}_t^k \in \mathbb{N}^{\delta^k}$, where $b_t^{k,d} = j$ indicates that UE k has j packets with time-to-live of d frames in its buffer. In the uplink slots, each packet copy includes the UE buffer status in its header. The index of the first non-zero component of $\boldsymbol{b}_t^k$ is the head-of-buffer index of UE k. It is denoted by h_t^k. The matrix of the buffers of all UEs at t is denoted $\boldsymbol{B}_t \in \mathcal{M}_{K\times\delta}(\mathbb{N})$, where $\delta = \max_k \delta_k$ and the k-th line of $\boldsymbol{B}_t$ is defined as the vector $\boldsymbol{b}_t^k$ padded with $\delta - \delta_k$ zeros at its end. At t, if UE k is successfully decoded, the BS retrieves b_t^k and hence h_t^k. The BS then feeds back a K-dimensional vector $\boldsymbol{\alpha}_t$ such that $\alpha_t^k = 2^{\delta_k - h_t^k}/(2^{\delta_k} K)$ if the UE k has sent a successfully decoded packet, and

$\alpha_t^k = 0$ otherwise. This allows the UEs to both receive an ACK/NACK, and to evaluate the criticality of the successful transmissions. Indeed, the greater the criticality of the packet sent by UE k was, the greater the value α_t^k. The buffer statuses are updated at the end of each frame t following a buffer status transition function $\mathcal{T}_B \colon \boldsymbol{B}_t \mapsto \boldsymbol{B}_{t+1}$. This transition function consists in the following steps for each UE k: (a) Upon reception of the downlink feedback message, if the UE transmission was successful, b_t^{k,h_t^k} is decremented, i.e, $b_{t+1}^{k,h_t^k} = b_t^{k,h_t^k} - 1$; (b) The packets designated by $b_t^{k,1}$ expire, for $d \in [\![1, \delta_k - 1]\!]$, $b_{t+1}^{k,d} \leftarrow b_t^{k,d+1}$, and $b_{t+1}^{k,\delta^k} \leftarrow 0$; (c) New packets may have been generated during the frame but cannot be transmitted before the next frame. Denoting t_g^k the generation time of a new packet relative to the beginning of the frame, if $1 - t_g^k/T_f \leq \mathrm{frac}(\overline{\delta^k}/T_f)$, the new packet is being added to b_{t+1}^{k,δ_k}. Otherwise, it is being added to $b_{t+1}^{k,\delta_k - 1}$. We define $\boldsymbol{G}_t \in \mathcal{M}_{K,2}(\mathbb{N})$, the packet generation matrix, such that $G_t^{k,1}$ is equal to the number of packets generated by UE k at t with a time-to-live of $\delta^k - 1$, and $G_t^{k,2}$ is equal to the number of packets generated with a time-to-live of δ^k. We also introduce the estimated buffer matrix at time t as $\widehat{\boldsymbol{B}}_t \in \mathcal{M}_{K\times\delta}(\mathbb{N})$. This matrix is known by the BS, and is an estimation of the UEs' current buffer statuses. At frame 0, $\widehat{\boldsymbol{B}}_t = \boldsymbol{0}_{\mathcal{M}_{K\times\delta}(\mathbb{N})}$. At the end of each frame t, the k-th line of $\widehat{\boldsymbol{B}}_t$ is replaced by the k-th line of $\boldsymbol{B}_t$ if UE k was successfully decoded at t. Then, the steps (a) and (b) of the buffer status transition function $\mathcal{T}_B$ are applied to the estimated buffer matrix $\widehat{\boldsymbol{B}}_t$, yielding $\widehat{\boldsymbol{B}}_{t+1}$. Note that newly generated packets are not included in the estimation as the BS is not aware of their arrival in the buffers. The BS broadcasts a traffic vector $\widehat{\boldsymbol{b}}_{t+1} \in \mathbb{N}^\delta$ in the downlink feedback message. It is defined as $\widehat{\boldsymbol{b}}_{t+1} = \sum_{k=1}^K \widehat{\boldsymbol{B}}_{t+1}^{k,:}$. This allows the UEs to gain insight on the overall upcoming traffic, and on its criticality.

III. Problem formulation

Maximizing the system reliability under a strict latency constraint, we aim to minimize the number of lost packets from deadline expiration with respect to UE policies. This is equivalent to maximizing a URLLC score denoted by U, which we define over a communication period of H frames as the ratio between the total number of successfully decoded packets and the total number of generated packets, i.e,

$$U \coloneqq \frac{\mathbb{E}\left[\sum_{t=0}^{H-1}\sum_{k=1}^{K} \mathbb{1}_{\left\{\alpha_t^k \neq 0\right\}}\right]}{\mathbb{E}\left[\sum_{t=0}^{H-1}\sum_{k=1}^{K}\sum_{d=1}^{2} \boldsymbol{G}_t^{k,d}\right]}. \tag{1}$$

We can show that a sufficient condition to maximize U is to maximize $\mathbb{E}\left[\sum_{t=0}^{H-1}\sum_{k=1}^{K} \mathbb{1}_{\left\{\alpha_t^k \neq 0\right\}} + \alpha_t^k\right]$, a result used for the design of our reward. This maximization problem can be formulated as the planning problem of a Dec-POMDP, a fully-cooperative Markov game in which agents take actions depending on the observations they draw from their environment at discrete time intervals [24]. Following our system model, our Dec-POMDP is defined as follows:

- There are K agents, representing the UEs.
- The t-th frame begins at time step t.
- At time step t, the environment state is defined as $\boldsymbol{s}_t \coloneqq \boldsymbol{B}_t$. The initial state distribution ρ_0 is given by an initial packet generation step yielding $\boldsymbol{B}_0$.
- Each agent takes an action represented by $\boldsymbol{a}_t^k$, an m-dimensional vector corresponding to its slot choices, i.e, $a_t^{k,i} = 1$ if UE k transmits on the i-th slot and $a_t^{k,i} = 0$ otherwise. The agents' joint action is a $K \times m$ matrix denoted $\boldsymbol{A}_t$, which k-th line is defined as $\boldsymbol{a}_t^k$.
- Given $\boldsymbol{s}_t$, the joint action $\boldsymbol{A}_t$ yields a global reward r_t, which we define as

$$r_t = \sum_{k=1}^{K} \mathbb{1}_{\left\{\alpha_t^k \neq 0\right\}} + \alpha_t^k. \tag{2}$$

- The state transition function is defined as $\mathcal{T} \colon (\boldsymbol{s}_t, \boldsymbol{A}_t) \mapsto \mathcal{T}_B(\boldsymbol{s}_t)$. The stochastic nature of $\mathcal{T}$ solely depends on the traffic.
- The k-th agent's observation is defined as $\boldsymbol{o}_t^k(\boldsymbol{s}_t, \boldsymbol{A}_{t-1}) = \left(\boldsymbol{B}_t^{k,:}, \widehat{\boldsymbol{b}}_t, \boldsymbol{\alpha}_{t-1}\right)$ with the convention that $\boldsymbol{\alpha}_{-1} = \boldsymbol{0}_{\mathbb{R}^m}$.
- Similarly to the agent model approach described in [24], we define the internal state of an agent k at time t, $\boldsymbol{I}_t^k$, as the history of its past observations and actions: $\boldsymbol{I}_t^k \coloneqq \left(\boldsymbol{o}_t^k, \boldsymbol{a}_{t-1}^k, \boldsymbol{o}_{t-1}^k, \ldots, \boldsymbol{o}_1^k, \boldsymbol{a}_0^k, \boldsymbol{o}_0^k\right)$.

In the t-th frame, the k-th agent's policy, i.e, its slot choices behaviour, depends on its internal state, and is denoted $\pi^k\left(\cdot|\boldsymbol{I}_t^k\right)$. It is a probability distribution defined on the set $\{0,1\}^m$. UE k samples $\boldsymbol{a}_t^k$ according to this distribution. Given ρ_0, $\mathcal{T}$ and the joint policy $\boldsymbol{\pi}$, at time step t, $\boldsymbol{s}_t$ follows an on-policy distribution denoted ρ_π^t. Following a bias-variance tradeoff approach [25], the agents aim to find the joint policy $\boldsymbol{\pi}$ which maximizes the expected sum of discounted rewards:

$$J(\boldsymbol{\pi}) = \mathbb{E}_{\boldsymbol{s}_0\sim\rho_0, \boldsymbol{s}_{1:(H-1)}\sim\rho_\pi^{1:(H-1)}, \boldsymbol{A}_{0:(H-1)}\sim\pi}\left[\sum_{t=0}^{H-1}\gamma^t r_t\right],$$

where $\gamma \in [0,1)$ is the rewards' discount factor, allowing for a variance reduction of this objective's Monte-Carlo estimation at the cost of introducing bias. We hence aim to solve the following optimization problem:

$$\max_{\boldsymbol{\pi}} J(\boldsymbol{\pi}), \tag{3}$$

which we refer to as the uMRA problem.

IV. Multi-Agent Deep Reinforcement Learning

As a solution to the uMRA problem, we propose uMRA-HAPPO. It is described in detail in Algorithm 1. HAPPO [20] is an adaptation of the PPO algorithm [26] to the multi-agent setting. PPO is a single agent policy gradient algorithm. It relies on an approximated version of the concept of trust region introduced in the Trust Region Policy Optimization (TRPO) algorithm, TRPO being itself an heuristic approach derived from a policy iteration procedure which exhibits monotonical

Algorithm 1: uMRA-HAPPO

1 Initialize policy parameters $\boldsymbol{\theta}_0^1, \ldots, \boldsymbol{\theta}_0^K$ and the VF estimation parameters $\boldsymbol{\phi}_0$.
2 **for** *iteration* $n \leftarrow 0$ **to** $N-1$ **do**
3 The BS begins the data collection phase.
4 The UEs execute their current policy. Sample and memorize trajectories $\{(\boldsymbol{o}_{b,t}^k, \boldsymbol{a}_{b,t}^k, r_{b,t}, \boldsymbol{o}_{b,t+1}^k) \mid b \in [\![1,B]\!],\ t \in [\![0,H-1]\!],\ k \in [\![1,K]\!]\}$ along with the agents' internal states $\boldsymbol{I}_{b,t}^k$.
5 The BS begins the training phase.
6 The BS harvests the memorized trajectories from the UEs and reconstructs the sampled states $\boldsymbol{s}_{b,0}, \boldsymbol{s}_{b,1}, \ldots, \boldsymbol{s}_{b,H}$. The BS also computes the rewards-to-go $\widehat{R}_{b,t} = \sum_{t'=t}^{H-1} \gamma^{t'-t} r_{b,t'}$.
7 VF parameters initialization: $\boldsymbol{\phi}_{n+1} \leftarrow \boldsymbol{\phi}_n$.
8 **for** *epoch* $e \leftarrow 1$ **to** E **do**
9 At the BS, a random permutation of the UEs indices $(k_1, \ldots, k_K)$ is sampled.
10 $M^{k_1}(\boldsymbol{s}_{b,t}, \boldsymbol{A}_{b,t}) \leftarrow \widehat{A}(\boldsymbol{s}_{b,t}, \boldsymbol{A}_{b,t})$ for all $b \in [\![1,B]\!],\ t \in [\![0,H-1]\!]$, $\widehat{A}$ being the GAE estimator defined according to (6).
11 **for** $i \leftarrow 1$ **to** K **do**
12 UE k_i receives from the BS $\left(M^{k_{1:i}}(\boldsymbol{s}_{b,t}, \boldsymbol{A}_{b,t})\right)_{b \in [\![1,B]\!], t \in [\![0,H-1]\!]}$.
13 UE k_i updates $\boldsymbol{\theta}_n^{k_i}$ according to (4).
14 **if** $i < K$ **then**
15 UE k_i computes the values $M^{k_{1:i+1}}(\boldsymbol{s}_{b,t}, \boldsymbol{A}_{b,t})$ according to (5) and sends them to the BS, which passes them to UE k_{i+1}.
16 The BS updates the VF estimation with the Adam algorithm according to (7).

performance improvement guarantees in [27]. In the same way, HAPPO is an approximation of HATRPO, which is derived from a multi-agent policy iteration procedure with monotonical performance improvement guarantees in [20]. In HAPPO, the agents are heterogeneous, i.e, the policy of each agent k depends on an independent set of parameters $\boldsymbol{\theta}^k$. This is important in the uMRA context, as having distinct UE policies allows for a reduction of collision likelihood. It is also important as we can consider heterogeneous latency requirements among the UEs.

HAPPO is an iterative optimization process which alternates between data collection and training phases for N iterations (line 2). During the data collection phase of an iteration n, the agents, implemented in the UEs, take their actions according to their current parametrized policy $\pi_{\theta_n^k}^k$ (we lighten the notation by denoting $\pi_{\theta_n}^k := \pi_{\theta_n^k}^k$), and a batch of B trajectories of H time steps is sampled (line 4). At each time step, the UEs save their current observation, reward, and last taken action in a memory replay buffer. After the last time step, the BS sends a downlink message indicating that the training phase begins (line 5). Then, the BS harvests the memory replay buffers from the UEs and reconstructs the sequence of states encountered during the data collection phase (line 6), after which follows a sequential update procedure orchestrated by the BS, see Fig. 1 and lines 9-15. The sampled trajectories are used by each UE k_i to optimize its policy $\pi_{\theta_{n+1}^{k_i}}^{k_i}$ (line 13) in order to maximize the HAPPO surrogate objective [20] using the Adam algorithm [28]:

$$\boldsymbol{\theta}_{n+1}^{k_i} = \arg\max_{\boldsymbol{\theta}^{k_i}} \left[\frac{1}{BH} \sum_{b=1}^{B} \sum_{t=0}^{H-1} \min \Bigg(\frac{\pi_{\theta}^{k_i}\left(\boldsymbol{a}_{b,t}^{k_i} | \boldsymbol{I}_{b,t}^{k_i}\right)}{\pi_{\theta_n}^{k_i}\left(\boldsymbol{a}_{b,t}^{k_i} | \boldsymbol{I}_{b,t}^{k_i}\right)} M^{k_{1:i}}(\boldsymbol{s}_{b,t}, \boldsymbol{A}_{b,t}), \right. \\ \left. \text{clip}\left(\frac{\pi_{\theta}^{k_i}\left(\boldsymbol{a}_{b,t}^{k_i} | \boldsymbol{I}_{b,t}^{k_i}\right)}{\pi_{\theta_n}^{k_i}\left(\boldsymbol{a}_{b,t}^{k_i} | \boldsymbol{I}_{b,t}^{k_i}\right)}, 1 \pm \epsilon \right) M^{k_{1:i}}(\boldsymbol{s}_{b,t}, \boldsymbol{A}_{b,t}) \Bigg) \right], \tag{4}$$

where $\text{clip}(x, 1 \pm \epsilon)$ is equal to $1-\epsilon$ if $x \leq 1-\epsilon$, x if $1-\epsilon < x < 1+\epsilon$, and $1+\epsilon$ if $x \geq 1+\epsilon$. The coefficients $M^{k_{1:i}}(\boldsymbol{s}_{b,t}, \boldsymbol{A}_{b,t})$ are defined in [20] as:

$$\begin{aligned} M^{k_1}(\boldsymbol{s}_{b,t}, \boldsymbol{A}_{b,t}) &= \widehat{A}(\boldsymbol{s}_{b,t}, \boldsymbol{A}_{b,t}) \\ M^{k_{1:i+1}}(\boldsymbol{s}_{b,t}, \boldsymbol{A}_{b,t}) &= \frac{\pi_{\theta_{n+1}}^{k_i}\left(\boldsymbol{a}_{b,t}^{k_i} | \boldsymbol{I}_{b,t}^{k_i}\right)}{\pi_{\theta_n}^{k_i}\left(\boldsymbol{a}_{b,t}^{k_i} | \boldsymbol{I}_{b,t}^{k_i}\right)} M^{k_{1:i}}(\boldsymbol{s}_{b,t}, \boldsymbol{A}_{b,t}) \end{aligned} \tag{5}$$

for all $i \in [\![1, K-1]\!]$, $b \in [\![1,B]\!]$, $t \in [\![0, H-1]\!]$, $\widehat{A}(\boldsymbol{s}_{b,t}, \boldsymbol{A}_{b,t})$ being an estimation of the global advantage of taking the joint action $\boldsymbol{A}_{b,t}$ in state $\boldsymbol{s}_{b,t}$ (lines 10 and 15). Here, we use the generalized advantage estimator (GAE) defined in [25] as :

$$\widehat{A}(\boldsymbol{s}_{b,t}, \boldsymbol{A}_{b,t}) = \sum_{l=0}^{H-t-1} (\gamma \lambda_{GAE})^l \left(1 - \lambda_{GAE}^{H-t-l}\right) \delta_{b,t+l}^{V_{\phi_n}}, \tag{6}$$

where the coefficients $\delta_{b,t}^{V_{\phi_n}} = r_{b,t} + \gamma V_{\phi_n}(\boldsymbol{s}_{b,t+1}) - V_{\phi_n}(\boldsymbol{s}_{b,t})$ are the temporal difference residuals, V_{ϕ_n} being a parametrized estimation of the Value Function (VF) of the Dec-POMDP computed at the BS, and λ_{GAE} is another parameter to further adjust the bias-variance tradeoff of the GAE estimator.

For a given encountered state $\boldsymbol{s}$, the estimated value $V_{\phi_n}(\boldsymbol{s})$ is taken to be the output of a Deep Neural Network (DNN) called the critic. At each training epoch, after the policies updates, the BS updates the critic network parameters with the Adam algorithm using a Mean Squared Error (MSE) loss function (line 16):

$$\boldsymbol{\phi}_{n+1} = \arg\max_{\boldsymbol{\phi}} \left[\frac{1}{BH} \sum_{b=1}^{B} \sum_{t=0}^{H-1} \left(V_{\phi}(\boldsymbol{s}_{b,t}) - \widehat{R}_{b,t} \right)^2 \right]. \tag{7}$$

Likewise, for each agent k_i, the likelihood $\pi_{\theta_n}^{k_i}\left(\boldsymbol{a}^{k_i} | \boldsymbol{I}^{k_i}\right)$ is computed by UE k_i with the output of a DNN called an actor, which input is $\boldsymbol{I}^{k_i}$. For coherency reasons, here we present the policies as depending on the internal state of their corresponding agents. Indeed, even though the theoretical results

TABLE I
UMRA-HAPPO PARAMETERS.

Parameter	Value
Number of training iterations (N)	1000
Number of hidden DNN layers	3
Hidden layer size	K
Trajectories batch size (B)	10
Trajectory length (H)	$100/(\lambda T_f K)$
Discount factor (γ)	0.3
λ_{GAE}	0.95
Clip coefficient (ϵ)	0.1
Entropy coefficient (β)	10^{-6}
Actor learning rate	10^{-3}
Critic learning rate	10^{-3}
Number of training epochs per iteration (E)	4

associated to HAPPO are derived assuming full observability of the agents, as stated in [20], good results are still empirically obtained in partially observable environments using Recurrent Neural Network (RNN) layers at the beginning of the policies models. We use Gated Recurrent Units (GRU), which is a standard implementation [29].

V. Experiments

A. Simulation settings

The parameters of our traffic model are adopted from the factory automation use case of the 3GPP 5G NR specifications on URLLC [23] where we consider $\overline{\delta_k} = \overline{\delta} = 1$ ms for $k \in [\![1, K]\!]$. UE packet generation rates are all equal: $\lambda_k = \lambda$ for $k \in [\![1, K]\!]$. Every slot is made of 1 OFDM symbol for a subcarrier spacing of 30 kHz so that its time duration is equal to $T_s = 35.67$ μs [30]. The values related to uMRA-HAPPO parameters are listed in Table I.

B. Baselines

For the sake of comparison, we consider the following baselines with the same network model: (i) **Random**: At each frame, each UE with a non-empty buffer uniformly chooses at random on which slots to transmit. This algorithm is an Irregular Repetition Slotted Aloha (IRSA) with a uniform degree distribution; (ii) **Round Robin (RR)**: The UEs follow a fixed cyclic slot allocation. This scheme is similar to the Configured Grant Type 1 mechanism of 5G NR systems [9]; (iii) **Earliest Deadline First (EDF) - Oracle**: At every radio frame, the m most urgent head-of-buffer packets are scheduled without collision. This baseline is an oracle as it requires each UE to have full knowledge of all buffers.

C. Simulation results

In Fig. 3, we show the convergence of uMRA-HAPPO, uMRA-HAPPO without RNN, and uMRA-HAPPO without criticality and traffic feedback. We observe that the performance increases rapidly in all cases. Incorporating the feedback vectors broadcasted by the BS in the agent observations noticeably increases performance. Using an RNN increases convergence speed. In Fig. 4, 5, and 6, results were obtained by training uMRA-HAPPO on $N \times B \times H$ time steps, and then simulating runs generating 2×10^5 packets in average, for a total of $2 \times 10^5/(\lambda T_f K)$ time steps per run. In Fig. 4,

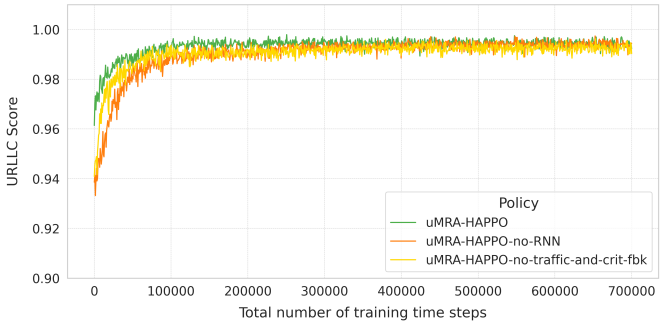


Fig. 3. Convergence for $K = 10$, $\lambda = 500$ s^{-1}, $m = 6$.

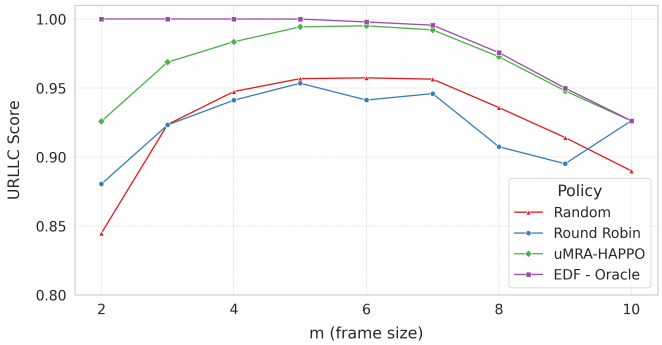


Fig. 4. Frame size tradeoff for $K = 10$, $\lambda = 500$ s^{-1}.

we show the URLLC score as a function of the frame size m for uMRA-HAPPO and the baselines. uMRA-HAPPO outperforms the realistic baselines and approaches the performance of the EDF oracle when m increases. We also observe a tradeoff for all schemes but RR: if m is too low, there are too many collisions that degrade the performance; if m is too high, there are less retransmission opportunities across radio frames and the system is more vulnerable to traffic peaks. For RR, the UEs are scheduled without collisions, there is thus no retransmissions. When m increases, the protocol overhead (D and G slots) decreases, which improves the performance. Some combinations of $\overline{\delta}$, m and K may however lead to lower scores (as here for $m = 8$ or 9) because of the late arrival of the ACK. In all cases, RR resource allocation is inefficient in presence of sporadic traffic. In Fig. 5 and 6, we use optimal frame sizes and show the URLLC scores as a function of K and λ, respectively. We observe that uMRA-HAPPO outperforms the realistic baselines, especially after $K = 6$ users and is close to the oracle performance up to $K = 10$. It is also the case up to $\lambda = 500$ s^{-1} when $K = 10$.

VI. Conclusion

We propose a MARL approach to solve the uplink MA problem for URLLC in a MRA environment. Reformulating our uMRA problem of system reliability maximization under strict latency constraints into the Dec-POMDP framework, we leverage the HAPPO algorithm to propose uMRA-HAPPO, numerically approximating a solution to the uMRA problem. We assess its performance and superiority on two baselines:

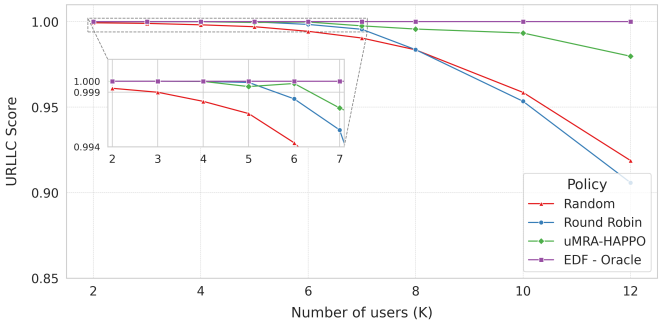


Fig. 5. Scores for $\lambda = 500$ s^{-1}.

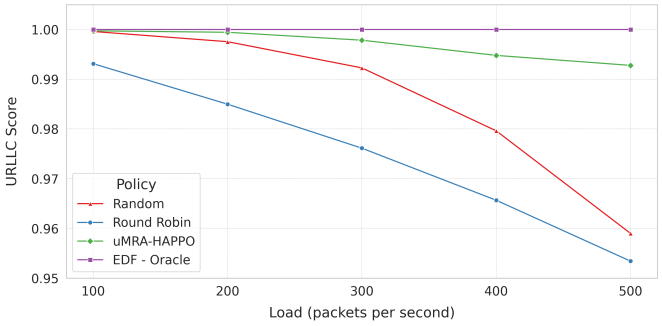


Fig. 6. Scores for $K = 10$.

a standard MRA solution, and a 5G NR inspired periodic resource allocation scheme. We highlight a tradeoff in the radio frame size in a URLLC context: When increasing, the number of collisions decreases but the number of retransmissions opportunities also decreases. Future work can involve extending uMRA-HAPPO to more realistic scenarios, e.g. with time-varying channels and generalization to time-frequency resource block allocations. Other scenarios could also include variable numbers of users in the network.

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